Harsh Harsh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.34
JNV GULABGARH YAMUNANAGAR	Class XII (CBSE)	2022	93.20%
JNV GULABGARH YAMUNANAGAR	Class X (CBSE)	2020	94.00%

SKILLS

Languages: C, C++, Python, TypeScript, SQL, HTML, CSS

Backend & APIs: Node.js, Express.js, Fastify, FastAPI, REST APIs

Frontend: Vite, HTML5, CSS3

Databases: PostgreSQL, Prisma ORM, Redis

Testing: Vitest, Jest, Supertest

Machine Learning: PyTorch, CNNs, Librosa, NumPy, Matplotlib

Tools: Git, GitHub, Docker, VS Code, Postman, Render, Vercel, k6

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems,Computer Network

PROJECTS

Payment Orchestration & Wallet Platform | TypeScript, Node.js, Prisma

- Engineered a modular payment backend supporting merchant/customer authentication, wallet payments, gateway flows, hybrid payments, and order tracking.

- Secured payment and refund operations with idempotency keys and Prisma transactions, preventing duplicate processing and conflicting key reuse during retries.

- Implemented reserve-release-finalize wallet debits, split refunds, and gateway fallback for insufficient balances with consistent transactional state management.

- Added Jest/Supertest integration tests, k6 load testing, and Dockerized PostgreSQL/Redis infrastructure to validate reliability and backend performance.

C++ Key-Value Store | C++, Crow

- Engineered a multithreaded in-memory key-value store with a Crow REST API supporting SET, GET, DELETE, EXISTS, and TTL-based key expiration.

- Implemented O(1) hash-table lookups and LRU cache eviction using unordered_map and doubly linked lists for efficient access and memory management.

- Added append-only persistence for recovering key-value data across restarts while separating storage, caching, and persistence into modular components.

- Designed an extensible OOP architecture with command handling and modular storage layers, then Dockerized the service for reproducible deployment.

Music Emotion Recognition | PyTorch, Librosa

- Developed a CNN-based regression model using PyTorch and Mel-spectrogram features to predict continuous valence and arousal from the DEAM dataset.

- Built an end-to-end ML pipeline with custom Dataset/DataLoader components, feature normalization, model training, evaluation, and checkpoint saving.

- Designed three Conv2D blocks with BatchNorm, ReLU, MaxPooling, adaptive average pooling, and a bounded regression head for emotion prediction.

- Achieved 0.23 RMSE and 0.19 MAE on the normalized test set, corresponding to less than 1 unit error on the original DEAM scale.

Page Pulse - Website Auditor | Fastify, Vite

- Built a full-stack website auditing platform analyzing SEO, metadata, accessibility, HTTP status, response time, and content metrics for public webpages.

- Designed a modular Fastify backend with Zod validation and centralized error handling to provide consistent and reliable API responses.

- Implemented automated API testing with Vitest and structured audit metrics to generate actionable reports from analyzed webpage content.

- Deployed the frontend on Vercel and backend on Render, integrating the complete website-auditing workflow into a reproducible application.

EXTRACURRICULAR

Participated in two **NCC** camp **ATC I,II**

A certificate holder in **NCC**

Silver Medalist, Fresher Basketball Tournament, IIT Hyderabad (2023)

NSS Volunteer - Contributed to social welfare drives and awareness campaigns as part of the National Service Scheme.

RELEVANT COURSES

Mathematics(IITH curriculum) - Calculus, Linear Algebra, Differential Equations, Probability, Complex Variables

Computer Science - Introduction to Programming(C language)

Neural Networks and Deep Learning - DeepLearning.AI, Coursera

Structuring Machine Learning Projects - DeepLearning.AI, Coursera